Neuron Performance Report

This report summarizes the performance of different neurons based on key metrics.

Metrics include KS Statistic, Pearson Correlation, and RMSE.

The Overall Score is a weighted combination of these metrics.

Metadata

Ground Truth File: data/cGround_Truth_data_10trials_150stims_180000ms.hdf5

Submission File: data/cNo_Inhibitory_Neurons_10trials_150stims_180000ms.hdf5

Number of Ground Truth Trials: 10

Number of Submission Trials: 10

Number of Neurons in Ground Truth: 5

Number of Neurons in Submission: 5

Number of Bins in PSTH: 360

Bin Size for PSTH: 500 ms

Kernel Size for Median Filter: 101

This report summarizes the performance of different neurons based on key metrics.

The metrics include KS Statistic, Pearson Correlation, and RMSE.

KS Statistic is...

A perfect Pearson correlation is 1. A correlation of 0 indicates no linear relationship.

RMSE is the square root of the average of squared differences between predicted and actual values. A 0 RMSE indicates no err

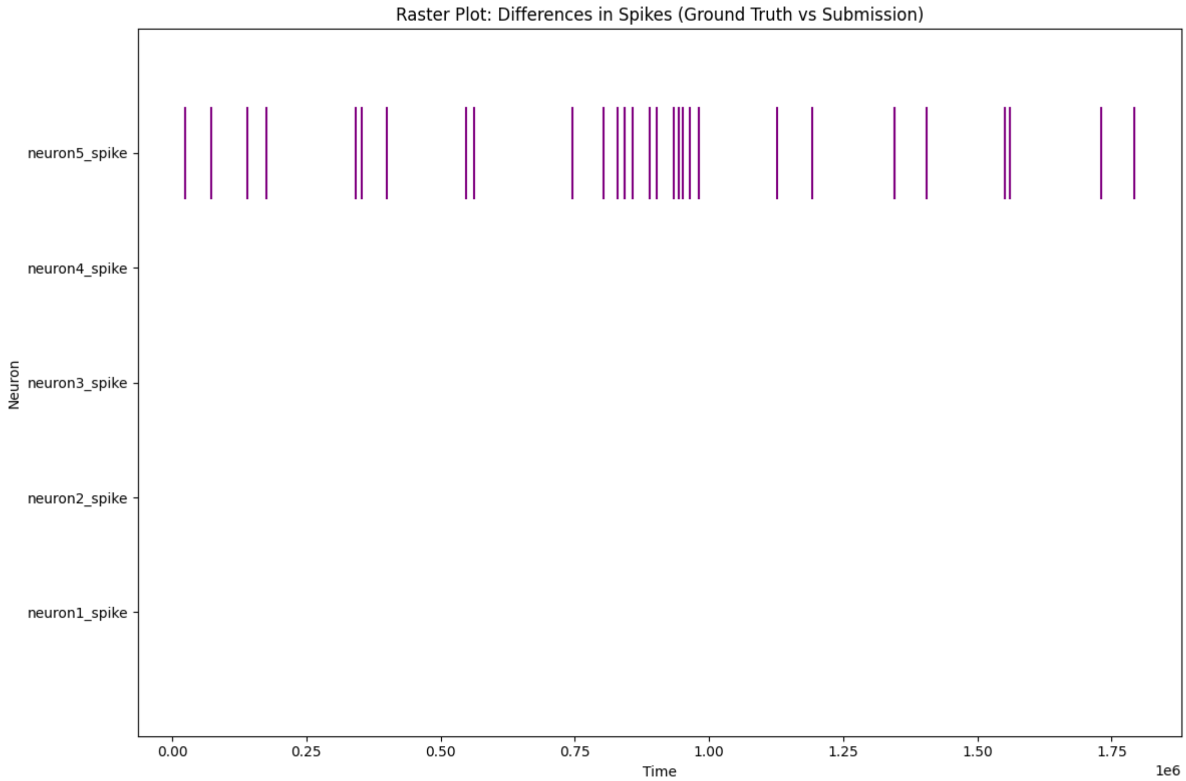
The Overall Score is a weighted combination of these metrics.

Summary Table of Spike Train Metrics

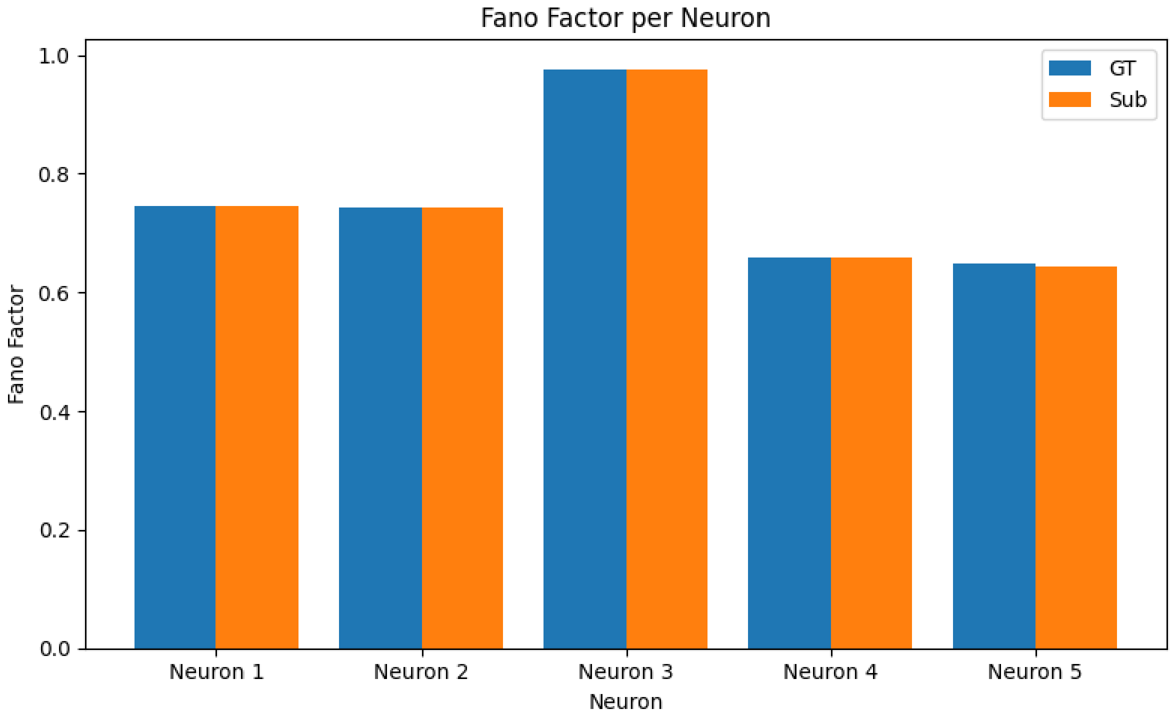
Neuron	KS Statistic	Pearson Correlation	RMSE	Overall Score
Neuron 1	0.0	1.0	0.0	100.0
Neuron 2	0.0	1.0	0.0	100.0
Neuron 3	0.0	1.0	0.0	100.0
Neuron 4	0.0	1.0	0.0	100.0
Neuron 5	1.389e-05	0.9934	5.676e-05	0.0

Raster Plot Comparison

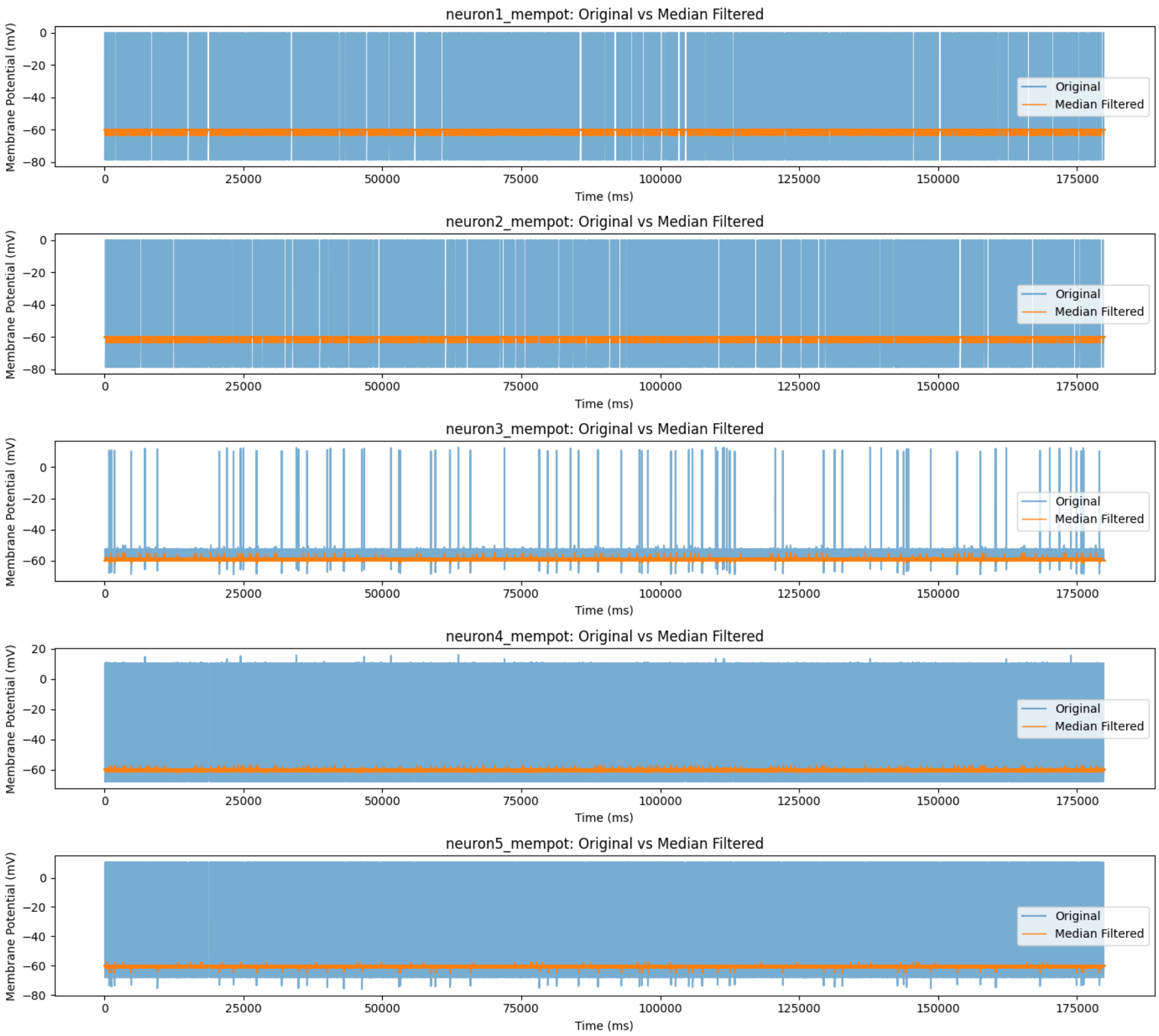
A raster plot is a fundamental visualization tool in neuroscience for representing spike trains. It displays neural activity across multiple trials or neurons in a compact and intuitive way. In a raster plot, time is typically on the x-axis, and each horizontal line (or row) represents a single trial or a single neuron. A small mark, often a vertical line or a dot, is placed on each row at the precise time a spike occurred. By stacking these rows together, a raster plot allows researchers to easily visualize patterns in spiking activity, such as changes in firing rate over time (e.g., in response to a stimulus), the trial-to-trial variability of a neuron's response, or the synchronized firing of multiple neurons.



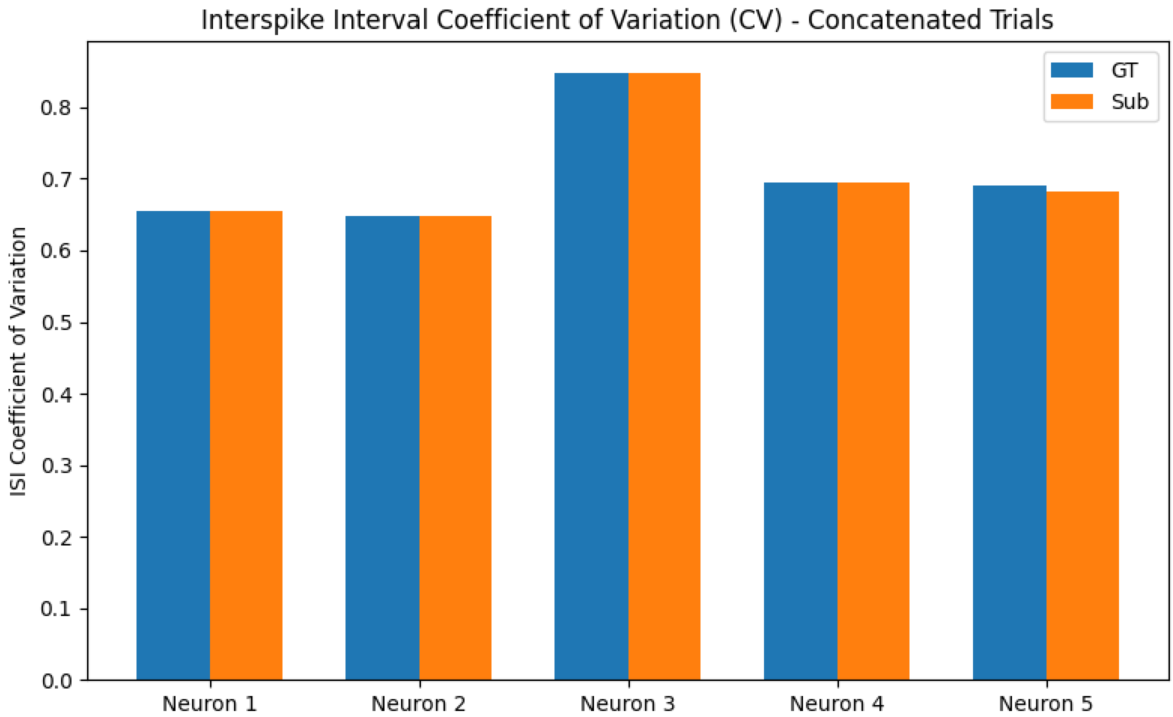
fano_factor.png



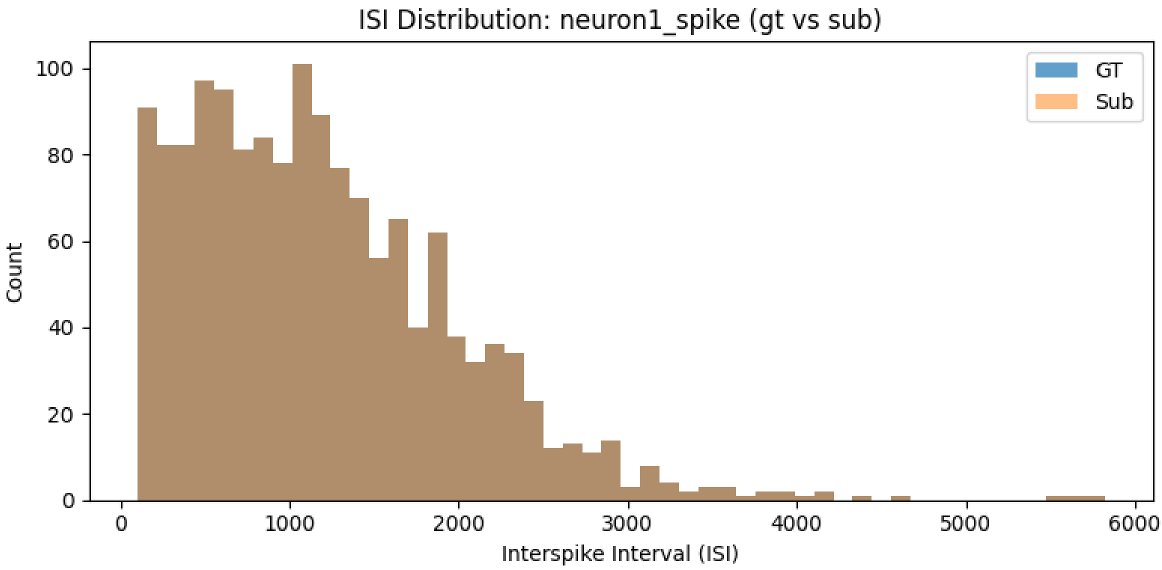
gt_filtered_mempot.png



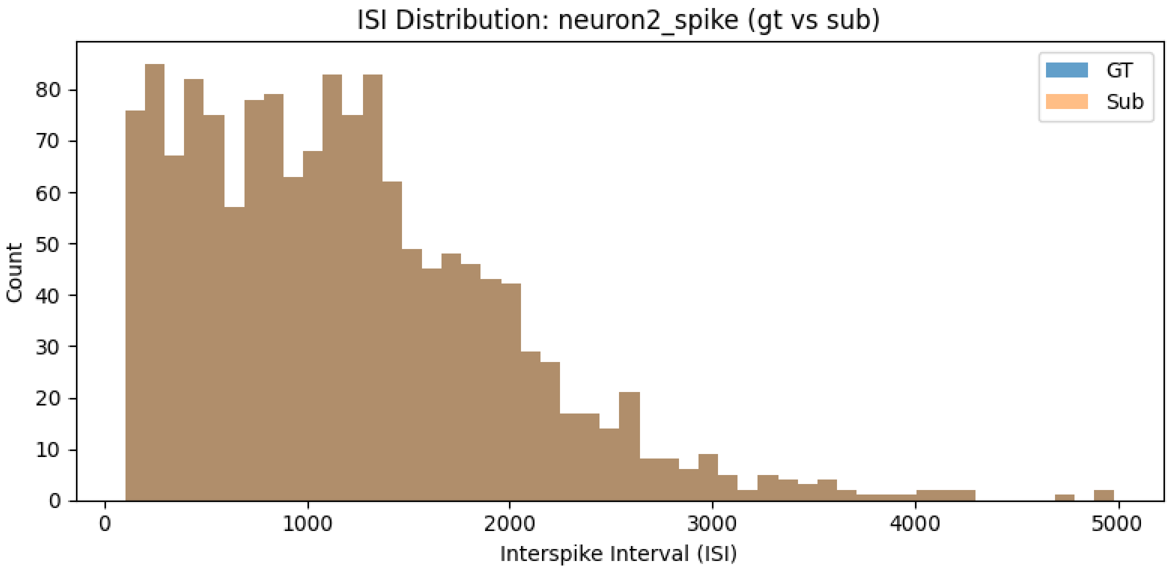
isi_cv.png


Interspike Interval Coefficient of Variation (CV) - Concatenated Trials
GT
Sub
ISI Coefficient of Variation
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Neuron 1
Neuron 2
Neuron 3
Neuron 4
Neuron 5

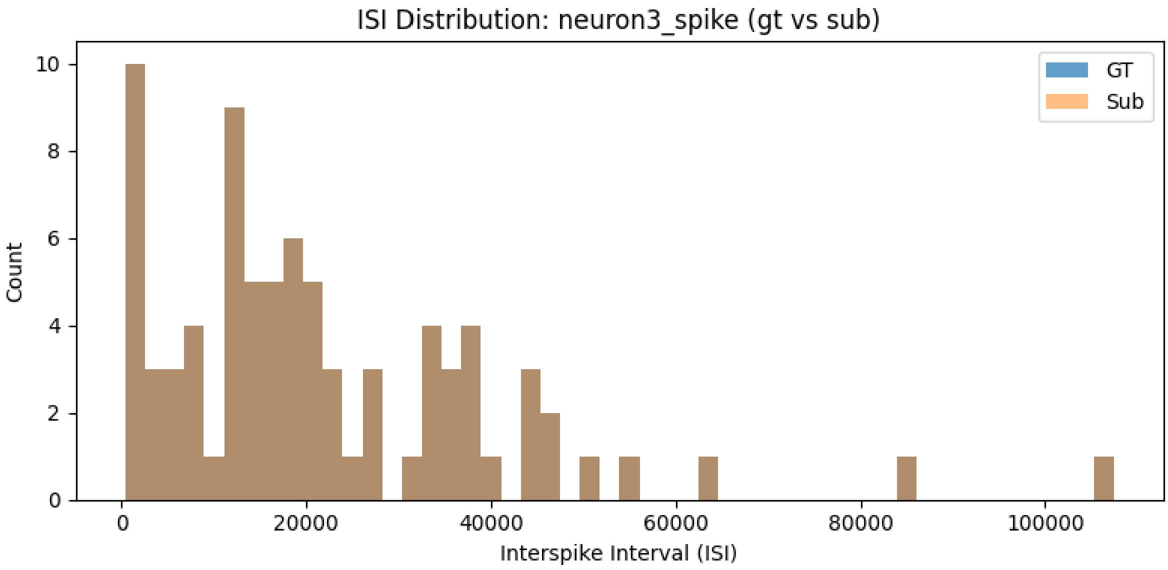
isi_dist_neuron1.png



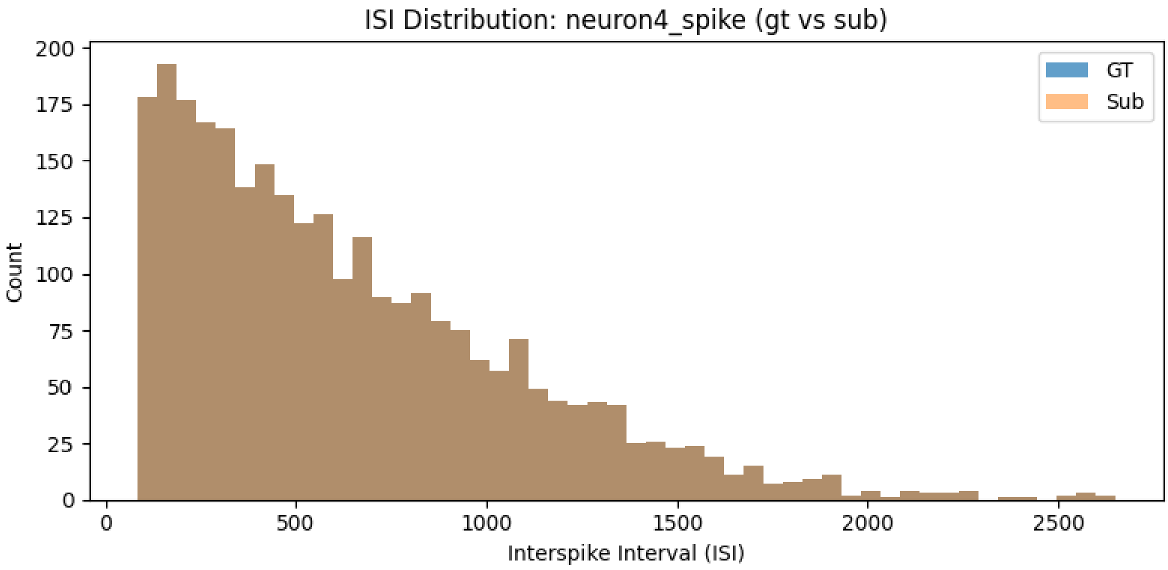
isi_dist_neuron2.png



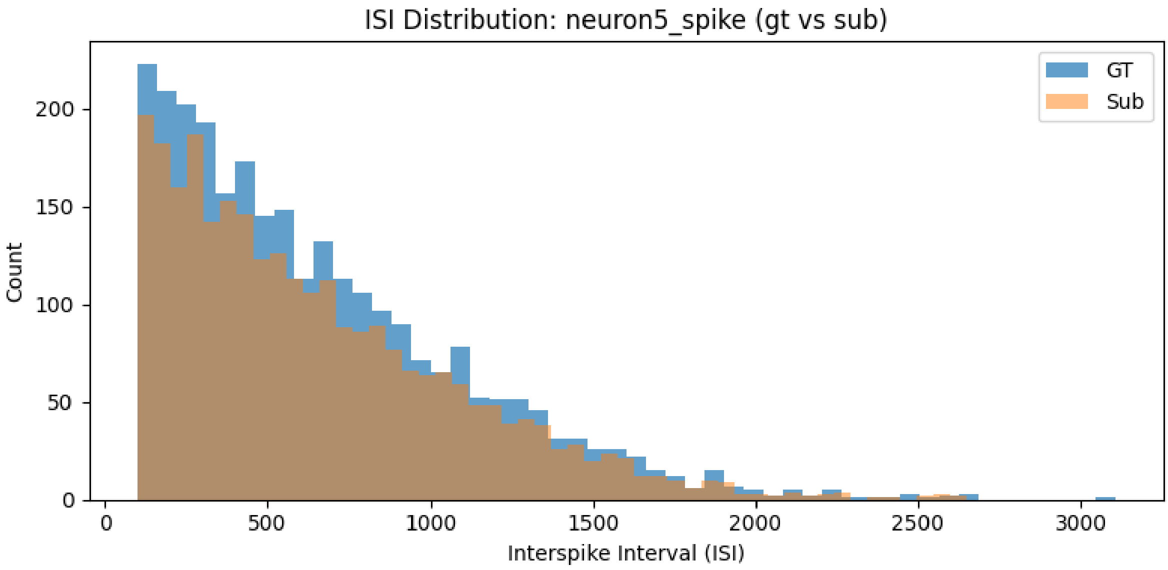
isi_dist_neuron3.png



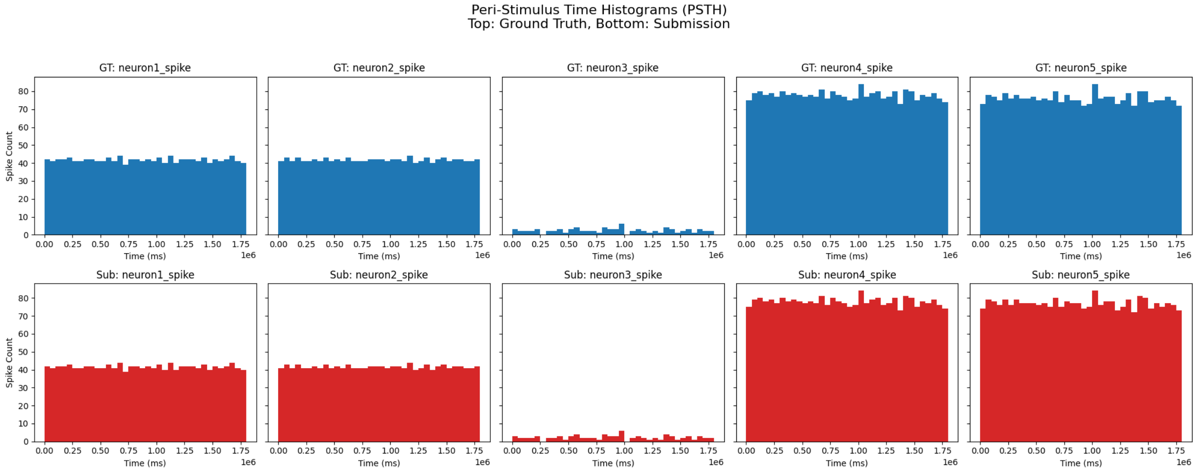
isi_dist_neuron4.png



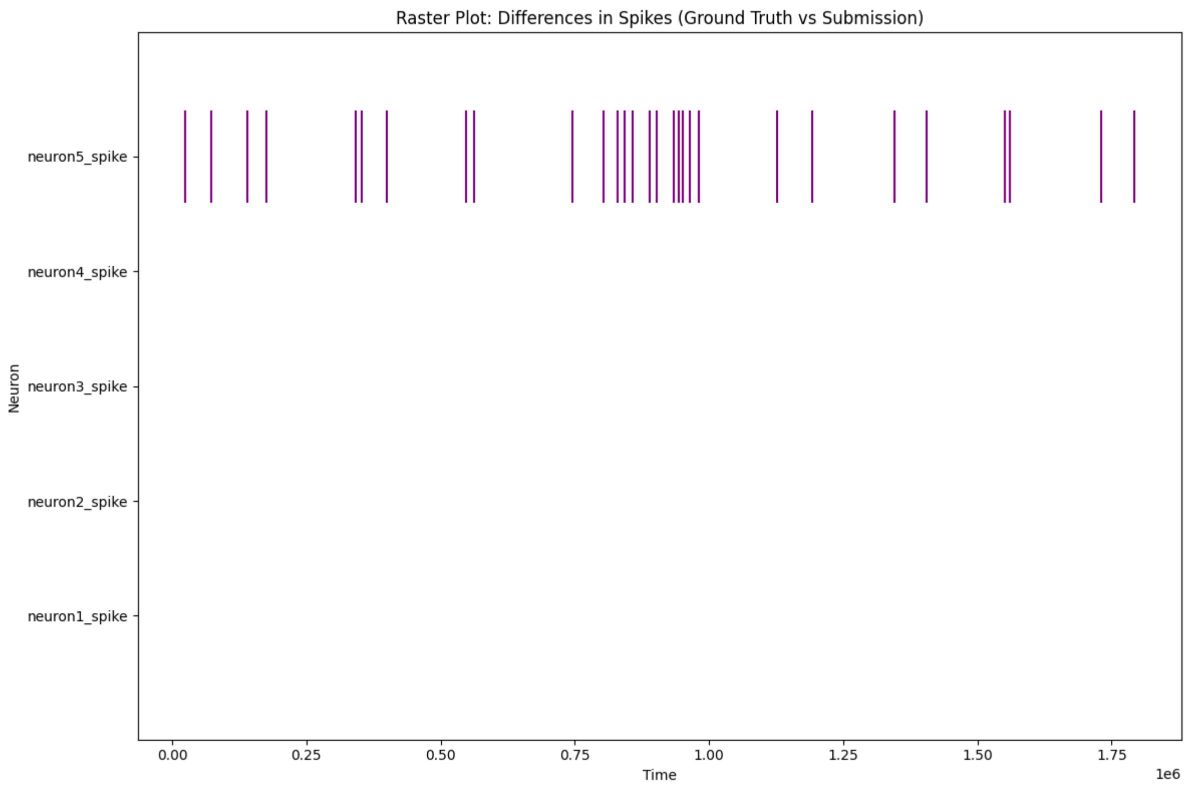
isi_dist_neuron5.png



peri_stim_histogram.png



raster_difference.png



sub_filtered_mempot.png